Understanding Beyond Experience

Learning to Verify and Revise World Understanding

Tianjun Gu | Fall 2027 PhD Research Interests

How can an embodied agent build an understanding of the world that goes beyond its own experience?

An embodied agent must learn from experiences that are incomplete and chosen for particular goals. Its world model may predict a plausible future while overlooking the physical condition that makes that future possible. I propose to study how agents actively test these conditions and accumulate knowledge that remains useful as their goals change. The central challenge is to connect verification to experience acquisition: a remembered uncertainty should suggest what to observe, and the resulting evidence should improve both memory and future predictions.

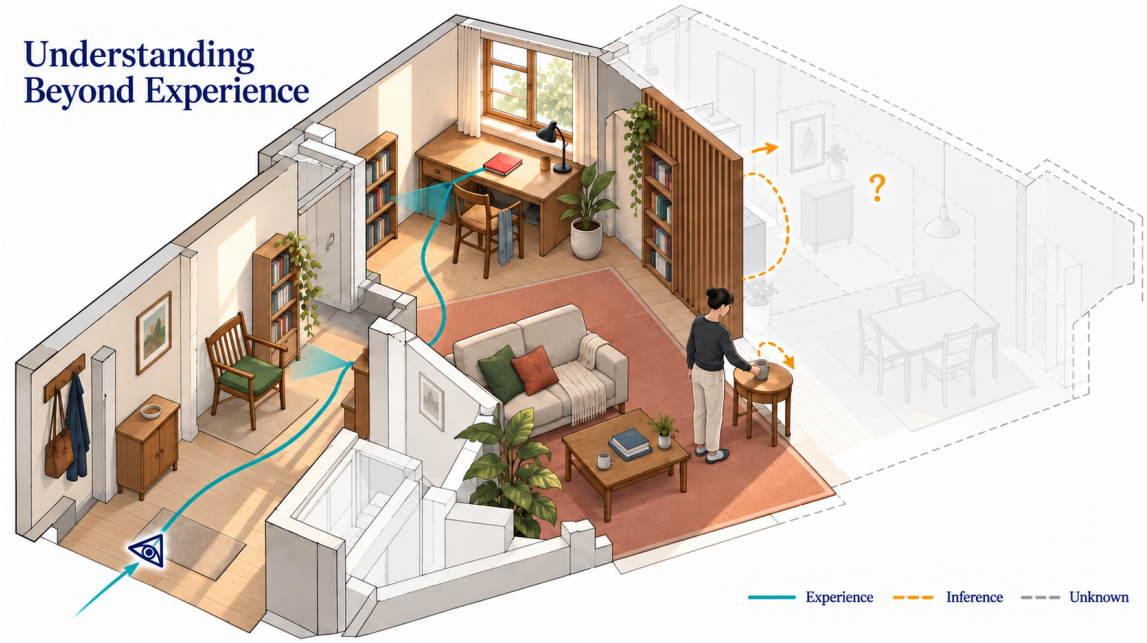


Figure 1. Selective experience as a foundation for reusable world understanding.

Research foundations

My first-author research studies the physical and temporal foundations of world understanding. My work in revision at Nature Communications introduces Teleo-Spatial Intelligence and EscherVerse, revealing gaps across 27 VLMs in reasoning about physical dynamics and intent [1]. My ECCV 2026 paper introduces VSI-Super-Wild, following Cambrian-S to evaluate continual agent-object-environment reasoning across 284.52 hours of real-world video [2].

My self-evolving spatial agent studies evidence retrieval, contradiction detection, and local belief updates [3]. DORAEMON connects hierarchical memory to safety-aware navigation planning [4]. Both are first-author manuscripts under review. At AI4CE, my work on active spatial supersensing asks how memory guides observation and supports future tasks. I would extend this toward agents that recognize when their current account of the world needs an external check.

Research directions

RoboMME exposes the demands of history-dependent robot behavior [5]. World Action Verifier separates state plausibility from action reachability to support world-model self-improvement [6]. I propose to connect these ideas through persistent memory of what was verified, what remains uncertain, and which observation could resolve it.

1. Remember the basis of a prediction

I will explore world memories that associate spatial and physical claims with their supporting observations and conditions of validity. Such a memory would preserve alternative explanations when the same visible outcome admits different causes. It should enable the agent to locate the assumption behind a prediction, rather than simply retrieve a similar past episode.

2. Acquire evidence that can overturn a belief

I will study how disagreements between remembered relations and predicted outcomes guide observation. A new view or a limited interaction would be chosen for its ability to distinguish plausible explanations. The learning question is how to reward useful correction, including evidence that rejects a confident prediction, while accounting for the cost of obtaining it.

3. Let verified experience support new goals

I will investigate how a correction acquired during one task changes later predictions and decisions. The agent should retain the relation established by an experience, not only a successful trajectory. When the context changes, memory would help identify whether that relation still applies or needs renewed verification.

Evaluation

Evaluation will test whether selective verification improves later task success and reduces repeated errors under matched sensing and memory budgets. Initial exploration will not reveal future test goals. I will compare retrieval-only memory, fixed observation strategies, and verification without persistent memory, assessing same-environment reuse separately from transfer to unseen scenes.

Selected references

[1] T. Gu et al. Vision-language models lag human performance on physical dynamics and intent reasoning. arXiv:2601.01547, 2026. Source

[2] T. Gu et al. Towards Spatial Supersensing in the Wild. ECCV, 2026. Project and benchmark materials. Source

[3] T. Gu et al. Self-Evolving Spatial Belief for Video Spatial Reasoning. Manuscript under review; available upon request.

[4] T. Gu et al. DORAEMON: Decentralized Ontology-aware Reliable Agent with Enhanced Memory Oriented Navigation. arXiv:2505.21969, 2025; manuscript under review. Source

[5] Y. Dai et al. RoboMME: Benchmarking and Understanding Memory for Robotic Generalist Policies. ICML, 2026. Source

[6] Y. Liu et al. World Action Verifier: Self-Improving World Models via Forward-Inverse Asymmetry. NeurIPS, 2026. Source